Sales & Customer Intelligence

SQL Server and Power BI portfolio case study

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A sales model built for business questions

Business questions

Where did revenue come from? Which customers remained active? How did product and channel margins compare?

My contribution

Built the SQL analytics layer, transformed data in Power Query, and developed a Power BI model and DAX measures.

1 fact table

Sales order detail grain

5 dimensions

Date, customer, product, territory and salesperson

55 measures

Reusable KPI definitions

The analytical scope

121,317

sales-detail rows

31,465

distinct orders

19,119

purchasing customers

109.85M

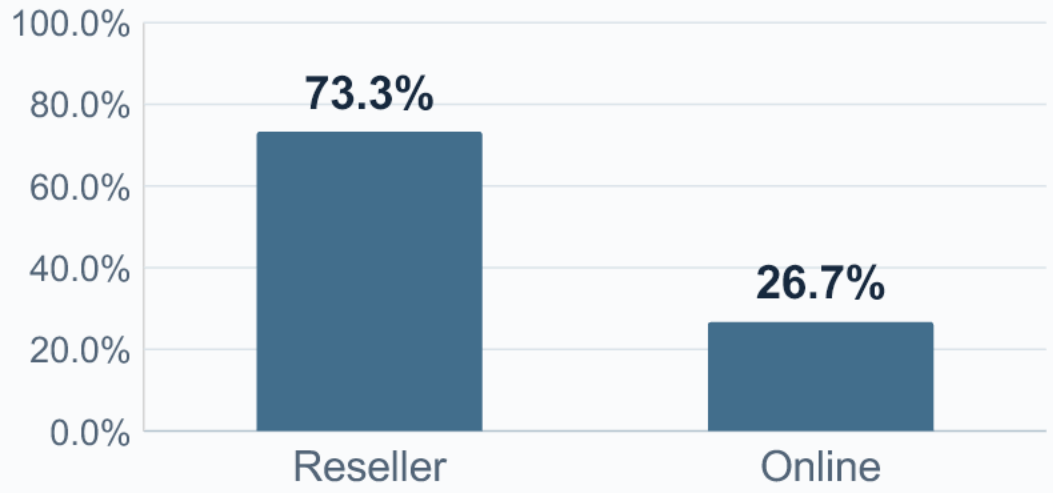
net sales in source monetary units

Reporting period: May 30, 2022 to June 29, 2025

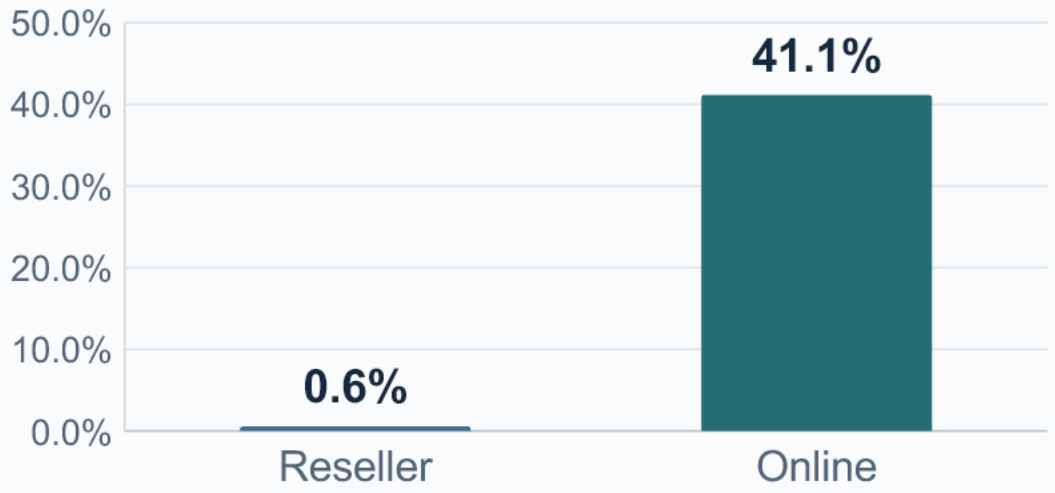
FINDING 01

Channel revenue and margin diverged

Share of net sales



Estimated standard-cost margin



Decision supported: review product mix, discounts and cost assumptions before changing channel strategy.

FINDING 02

Bikes accounted for 86.17% of sales



Revenue concentration

The category mix made dependence on bikes visible and gave users a starting point for product analysis.

A follow-up question

Accessories had a 49.87% estimated margin. Would a targeted cross-sell test be worthwhile?

Purchasing activity needed a clear definition

5,111

active within 90 days

14,008

inactive for more than 90 days

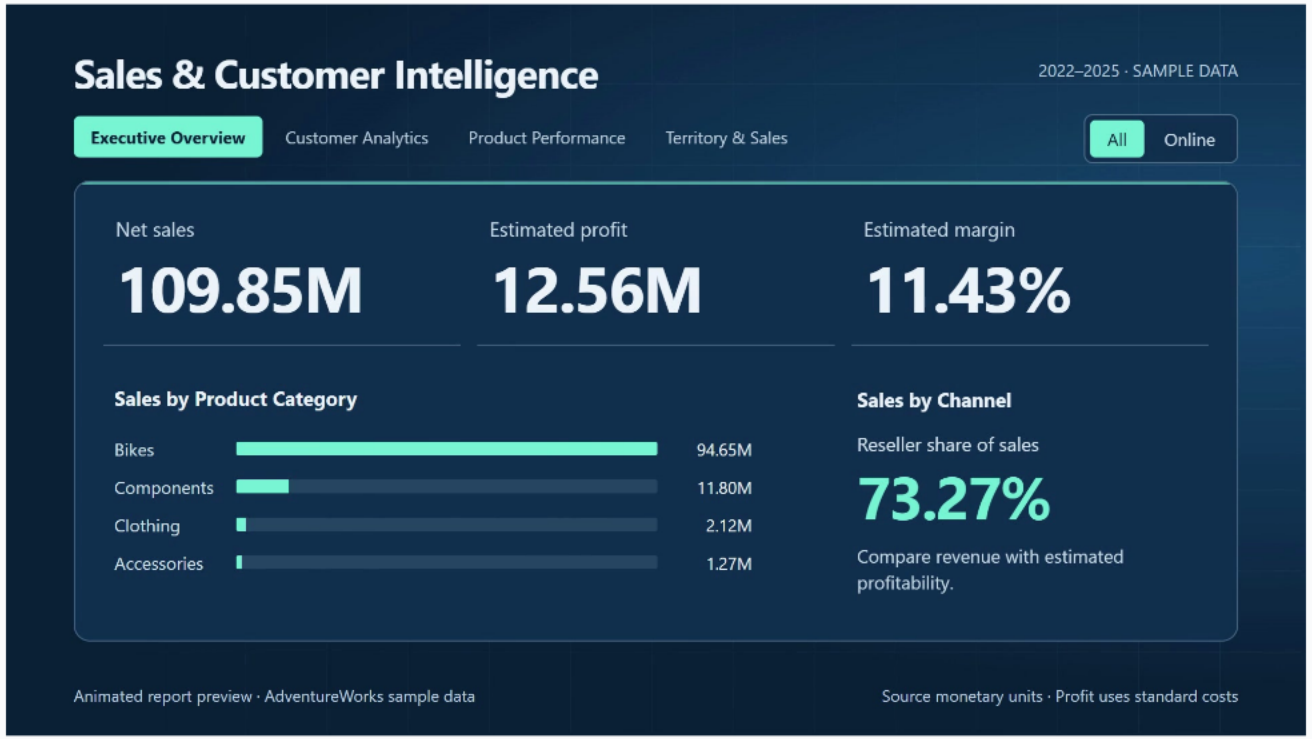
What the report enabled

Compared purchase recency with historical customer value to prioritize accounts for further investigation.

Interpretation

Measured activity as of June 29, 2025. The 90-day rule described recency, not a validated churn prediction.

Four report pages supported exploration



Executive Overview
Customer Analytics
Product Performance
Territory & Sales Performance

Shared filters, page navigation and metric definitions supported consistent exploration.

Watch the report walkthrough

Reliable reporting required validation

Data checks

Validated fact grain, dimension keys and relationship coverage. The embedded data had no duplicate detail IDs or unmatched dimension keys.

Important limits

Profit used standard costs. Partial years required comparable periods. Customer names were not unique account identifiers.

Skills demonstrated

T-SQL, dimensional modeling, Power Query, DAX filter context, KPI definitions and business interpretation.

Business value

Provided a consistent way to investigate sales concentration, customer activity and estimated profitability in sample data.